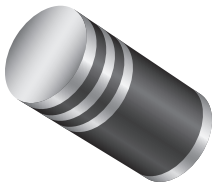
Surface Mount Schottky Barrier Rectifier



GL41 (DO-213AB)

FEATURES

- MELF Schottky rectifier
- Ideal for automated placement
- Guardring for overvoltage protection
- Low power losses, high efficiency
- Low forward voltage drop
- High surge capability
- Meets MSL level 1, per J-STD-020, LF maximum peak of 250 °C
- AEC-Q101 qualified
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	1.0 A
V_{RRM}	20 V,30 V, 40 V, 50 V, 60 V
I_{FSM}	30 A
V_F	0.50 V, 0.70 V
T_J max.	125 °C, 150 °C
Package	GL41 (DO-213AB)
Circuit configuration	Single

TYPICAL APPLICATIONS

For use in low voltage high frequency inverters, freewheeling, DC/DC converters, and polarity protection applications

MECHANICAL DATA

Case: GL41 (DO-213AB)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade
Base P/NHE3 - RoHS-compliant, AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test, HE3 suffix meets JESD 201 class 2 whisker test

Polarity: two bands indicate cathode end 1st band denotes device type 2nd band denotes voltage type

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER	SYMBOL	BYM13-20	BYM13-30	BYM13-40	BYM13-50	BYM13-60	UNIT
DENOTES SCHOTTKY DEVICES: 1st BAND IS ORANGE		**SGL41-20**	**SGL41-30**	**SGL41-40**	**SGL41-50**	**SGL41-60**	
Polarity color bands (2nd band) voltage type		Gray	Red	Orange	Yellow	Green	
Maximum repetitive peak reverse voltage	V_{RRM}	20	30	40	50	60	V
Maximum RMS voltage	V_{RMS}	14	21	28	35	42	V
Maximum DC blocking voltage	V_{DC}	20	30	40	50	60	V
Maximum average forward rectified current (fig. 1)	$I_{F(AV)}$	1.0					A
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	30					A
Voltage rate of change (rated V_R)	dV/dt	10 000					V/µs
Operating junction temperature range	T_J	-55 to +125			-55 to +150		°C
Storage temperature range	T_{STG}	-55 to +150					°C

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)									
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**BYM13-20 / SGL41-20**	**BYM13-30 / SGL41-30**	**BYM13-40 / SGL41-40**	**BYM13-50 / SGL41-50**	**BYM13-60 / SGL41-60**	**UNIT**
Maximum instantaneous forward voltage (1)	1.0 A		V_F	0.50	0.50	0.50	0.70	0.70	V
Maximum reverse current at rated DC blocking voltage (1)		T_A = 25 °C	I_R	0.5					mA
		T_A = 100 °C		10			5.0		
Typical junction capacitance	4.0 V, 1.0 MHz		C_J	110			80		pF

Note

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)							
PARAMETER	**SYMBOL**	**BYM13-20 / SGL41-20**	**BYM13-30 / SGL41-30**	**BYM13-40 / SGL41-40**	**BYM13-50 / SGL41-50**	**BYM13-60 / SGL41-60**	**UNIT**
Maximum thermal resistance (1)	$R_{\theta JA}$	75					°C/W
	$R_{\theta JT}$	30					

Note

(1) Thermal resistance junction to terminal, 0.24" x 0.24" (6.0 mm x 6.0 mm) copper pads to each terminal

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
SGL41-40-E3/96	0.137	96	1500	7" diameter plastic tape and reel
SGL41-40-E3/97	0.137	97	5000	13" diameter plastic tape and reel
BYM13-40-E3/96	0.137	96	1500	7" diameter plastic tape and reel
BYM13-40-E3/97	0.137	97	5000	13" diameter plastic tape and reel
SGL41-40HE3_A/H (1)	0.137	H	1500	7" diameter plastic tape and reel
SGL41-40HE3_A/I (1)	0.137	I	5000	13" diameter plastic tape and reel
BYM13-40HE3_A/H (1)	0.137	H	1500	7" diameter plastic tape and reel
BYM13-40HE3_A/I (1)	0.137	I	5000	13" diameter plastic tape and reel

Note

(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

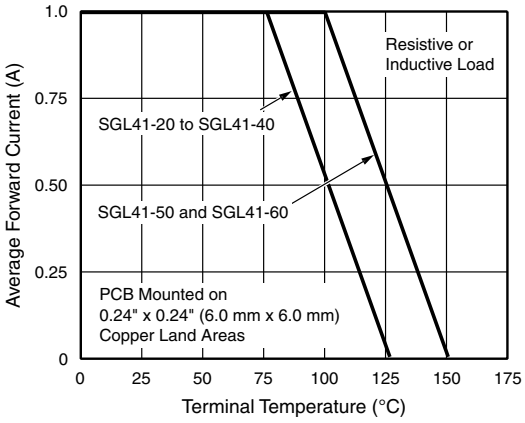


Fig. 1 - Forward Current Derating Curve

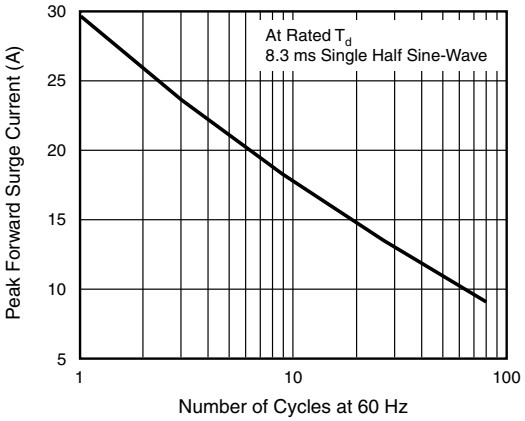


Fig. 2 - Maximum Non-Repetitive Peak Forward Surge Current

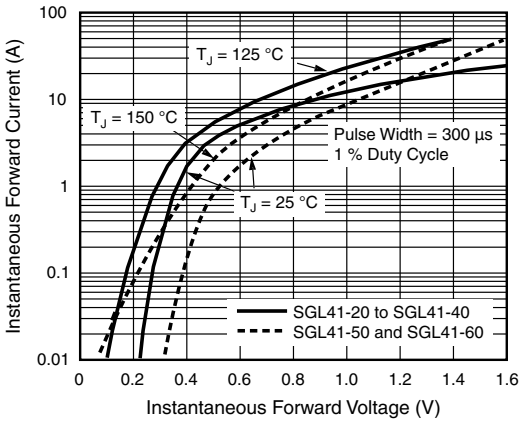


Fig. 3 - Typical Instantaneous Forward Characteristics

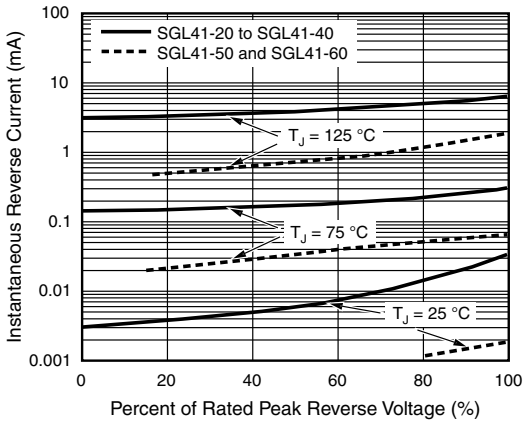


Fig. 4 - Typical Reverse Characteristics

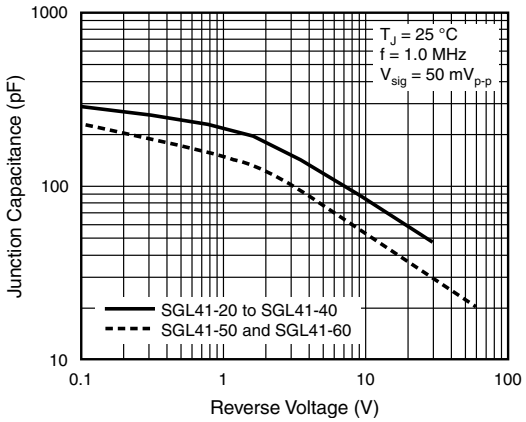
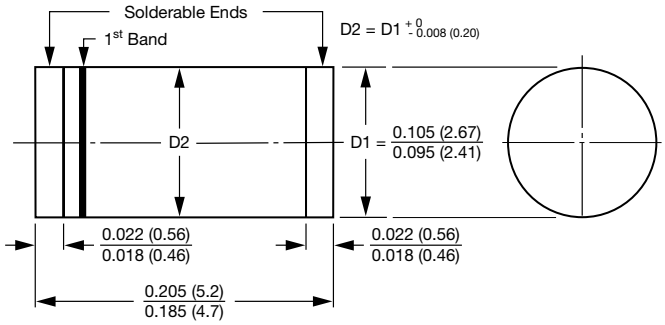


Fig. 5 - Typical Junction Capacitance

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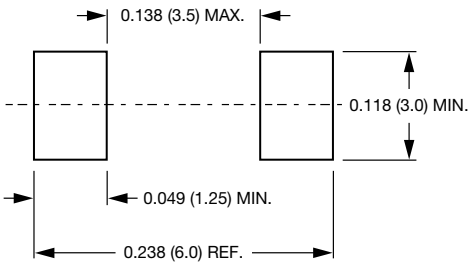
PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

GL41 (DO-213AB)



1st band denotes type and positive end (cathode)

Mounting Pad Layout



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